

Association between CYP2D6 polymorphisms and outcomes among women with early stage breast cancer treated with tamoxifen.

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Evidence abstract

CONTEXT The growth inhibitory effect of tamoxifen, which is used for the treatment of hormone receptor-positive breast cancer, is mediated by its metabolites, 4-hydroxytamoxifen and endoxifen.

The formation of active metabolites is catalyzed by the polymorphic cytochrome P450 2D6 (CYP2D6) enzyme.

OBJECTIVE To determine whether CYP2D6 variation is associated with clinical outcomes in women receiving adjuvant tamoxifen.

DESIGN, SETTING, AND PATIENTS Retrospective analysis of German and US cohorts of patients treated with adjuvant tamoxifen for early stage breast cancer.

The 1325 patients had diagnoses between 1986 and 2005 of stage I through III breast cancer and were mainly postmenopausal (95.4%).

Last follow-up was in December 2008; inclusion criteria were hormone receptor positivity, no metastatic disease at diagnosis, adjuvant tamoxifen therapy, and no chemotherapy.

DNA from tumor tissue or blood was genotyped for CYP2D6 variants associated with reduced (*10, *41) or absent (*3, *4, *5) enzyme activity.

Women were classified as having an extensive (n=609), heterozygous extensive/intermediate (n=637), or poor (n=79) CYP2D6 metabolism.

MAIN OUTCOME MEASURES Time to recurrence, event-free survival, disease-free survival, and overall survival.

RESULTS Median follow-up was 6.3 years.

At 9 years of follow-up, the recurrence rates were 14.9% for extensive metabolizers, 20.9% for heterozygous extensive/intermediate metabolizers, and 29.0% for poor metabolizers, and all-cause mortality rates were 16.7%, 18.0%, and 22.8%, respectively.

Compared with extensive metabolizers, there was a significantly increased risk of recurrence for heterozygous extensive/intermediate metabolizers (time to recurrence adjusted hazard ratio [HR], 1.40; 95% confidence interval [CI], 1.04-1.90) and for poor metabolizers (time to recurrence HR, 1.90; 95% CI, 1.10-3.28).

Compared with extensive metabolizers, those with decreased CYP2D6 activity (heterozygous extensive/intermediate and poor metabolism) had worse event-free survival (HR, 1.33; 95% CI, 1.06-1.68) and disease-free survival (HR, 1.29; 95% CI, 1.03-1.61), but there was no significant difference in overall survival (HR, 1.15; 95% CI, 0.88-1.51).

CONCLUSION Among women with breast cancer treated with tamoxifen, there was an association between CYP2D6 variation and clinical outcomes, such that the presence of 2 functional CYP2D6 alleles was associated with better clinical outcomes and the presence of nonfunctional or reduced-function alleles with worse outcomes.

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